

Dashboard Pages Documentation

Dashboard Overview

The Power BI dashboard is divided into four pages to provide a complete view of data quality, sales performance, profitability, and regional performance. The dashboard enables users to monitor data cleaning results and business insights through interactive visualizations and KPIs.

Page 1: Data Quality Overview

Purpose

This page focuses on evaluating the quality of the dataset before and after the cleaning process.

KPI Cards

Total Records

Displays the total number of records present in the raw dataset.

Total Records =
COUNTROWS('Raw Data')

Clean Records

Displays the number of records remaining after cleaning.

Clean Records =
COUNTROWS('Clean Data')

Missing Values Fixed

Shows the total number of missing values handled during cleaning.

Missing Values Fixed =
7

Duplicates Removed

Displays the total number of duplicate records removed.

Duplicates Removed =
10

Data Quality Score

Represents the overall quality of the cleaned dataset.

Data Quality Score =
98.5

Visualizations

Missing Values by Column

Visual Type: Clustered Bar Chart

Fields:

- Axis: Column Name
- Values: Missing Value Count

Purpose:
Identify columns with the highest number of missing values.

Data Issues Distribution

Visual Type: Donut Chart

Categories:

- Missing Values
- Duplicate Records
- Formatting Issues

Purpose:
Show distribution of data quality problems.

Before vs After Cleaning

Visual Type: Clustered Column Chart

Purpose:
Compare dataset quality before and after cleaning.

Data Quality Trend

Visual Type: Line Chart

Purpose:
Visualize improvement in data quality.

Page 2: Sales Performance Analysis

Purpose

Analyze sales performance across customers, regions, and time periods.

KPI Cards

Total Sales

Total Sales =
SUM('Clean Data'[Sales])

Average Sales

Average Sales =
AVERAGE('Clean Data'[Sales])

Maximum Sales

Maximum Sales =
MAX('Clean Data'[Sales])

Minimum Sales

Minimum Sales =
MIN('Clean Data'[Sales])

Visualizations

Sales by Region

Visual Type: Clustered Bar Chart

Fields:

- Axis: Region
- Values: Total Sales

Purpose:
Compare sales generated by each region.

Monthly Sales Trend

Visual Type: Line Chart

Fields:

- Axis: Month
- Values: Sales

Purpose:

Monitor sales growth over time.

Top 10 Customers by Sales

Visual Type: Horizontal Bar Chart

Fields:

- Axis: CustomerID
- Values: Sales

Purpose:

Identify highest-value customers.

Sales Distribution

Visual Type: Histogram

Purpose:

Analyze the spread of sales values.

Page 3: Profit Analysis

Purpose

Evaluate profitability across products, customers, and regions.

KPI Cards

Total Profit

Total Profit =
SUM('Clean Data'[Profit])

Average Profit

Average Profit =
AVERAGE('Clean Data'[Profit])

Profit Margin %

Profit Margin % =
DIVIDE([Total Profit],[Total Sales],0)

Loss Transactions

Loss Transactions =
CALCULATE(
COUNTROWS('Clean Data'),
'Clean Data'[Profit] < 0
)

Visualizations

Profit by Region

Visual Type: Clustered Column Chart

Purpose:
Compare profitability among regions.

Monthly Profit Trend

Visual Type: Line Chart

Purpose:
Track profit performance over time.

Profit vs Sales Relationship

Visual Type: Scatter Chart

Fields:

- X Axis: Sales
- Y Axis: Profit

Purpose:
Identify profitable and non-profitable transactions.

Profit Margin by Region

Visual Type: Bar Chart

Purpose:
Compare regional profit margins.

Page 4: Regional Performance Dashboard

Purpose

Provide a detailed view of regional business performance.

KPI Cards

Customer Count

Customer Count =
DISTINCTCOUNT('Clean Data'[CustomerID])

Best Region

Best Region Sales =
MAXX(
VALUES('Clean Data'[Region]),
[Total Sales]
)

Total Regional Sales

Regional Sales =
SUM('Clean Data'[Sales])

Total Regional Profit

Regional Profit =
SUM('Clean Data'[Profit])

Visualizations

Regional Sales Share

Visual Type: Donut Chart

Fields:

- Legend: Region
- Values: Sales

Purpose:
Display percentage contribution of each region.

Regional Profit Share

Visual Type: Donut Chart

Fields:

- Legend: Region
- Values: Profit

Purpose:
Show profit contribution by region.

Customer Distribution by Region

Visual Type: Tree Map

Fields:

- Group: Region
- Values: Customer Count

Purpose:
Understand customer concentration.

Regional Performance Matrix

Visual Type: Matrix

Columns:

- Region
- Total Sales
- Total Profit
- Customer Count
- Profit Margin %

Purpose:
Provide a comprehensive regional comparison.

Filters and Slicers Used

The following slicers are available on all dashboard pages:

Region Slicer

Allows filtering by:

- North
- South
- East
- West

Date Slicer

Allows filtering by:

- Year
- Quarter
- Month

Customer Slicer

Allows filtering by Customer ID.

Dashboard Navigation

Page	Description
Page 1	Data Quality Overview
Page 2	Sales Performance Analysis
Page 3	Profit Analysis
Page 4	Regional Performance Dashboard

Project Outcome

- Automated data cleaning process.
- Eliminated missing values and duplicate records.
- Improved data quality and consistency.
- Generated automated reports.
- Developed interactive Power BI dashboards.
- Reduced manual effort and reporting time.
- Enabled better business decision-making through data-driven insights.